

Indian Statistical Institute

BSDS II

Stochastic Process

Mid Term Sample Solutions

1. Consider a branching process starting with a single individual, $Z_0 = 1$ with the offspring distribution given by $\mathbb{P}(\xi = 0) = \mathbb{P}(\xi = 1) = \mathbb{P}(\xi = 2) = \mathbb{P}(\xi = 3) = 1/4$. Find $\mathbb{E}(Z_3)$ and $\text{Var}(Z_2)$. Also, find the probability that $\mathbb{P}(Z_3 = 0)$.

2. Let $\{\epsilon_n : n \geq 0\}$ be iid random variables taking values $\{-1, 1\}$ with probabilities $1/2$ each. Let $S_0 = 0$ and $S_n = \sum_{i=1}^n \epsilon_i$ for $n \geq 1$. Also, let $M_n = \max\{S_k : 0 \leq k \leq n\}$. Show that the sequence $\{Y_n = M_n - S_n : n \geq 0\}$ defines a Markov chain. Find the transition probabilities of this chain.

3. Let $\{\epsilon_n : n \geq 0\}$ be iid random variables taking values $\{-1, 1\}$ with probabilities $1/2$ each. Let $S_0 = 0$ and $S_n = \sum_{i=1}^n \epsilon_i$ for $n \geq 1$. Show that

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

4. Suppose a fair dice is being rolled. Let ϵ_n denote the face that turned up in the n -th roll. Let $M_n = \max\{\epsilon_k : 1 \leq k \leq n + 1\}$ for $n \geq 0$. Assume that $\{M_n : n \geq 0\}$ is a Markov chain on the state space $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. Compute $p_{2,5}^{(n)}, p_{33}^{(n)}$ and $p_{26}^{(n)}$.